

Test 24: Dirac string flux verification

Verdict: PASS | Generated: 2026-09-02 15:32:56 | Suite: AB-Cloud v19 (Julia 1.12.0)

What this test verifies

EXACT: plaquette flux = 2?q (mod 2?); tolerance = round-off accumulation across the lattice.

METHOD: EXACT: plaquette flux = 2?q (mod 2?); tolerance = round-off accumulation across the lattice.

Test 24 HARDCORE: 4-config full scan on 96x96, worst empty flux 8.88e-16, hermiticity true.

Result [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS

- 'plots/plot_01' (.png 600dpi / .svg / .pdf)

Plots

- 'plots/animation.gif' ? animated reveal of every plot of this test

Reproduce with: julia ab_cloud_v19.jl --test 24 --no-two-pass

Independent verification guide

Recompute over logs in logs/computation_log.txt (timestamps + deltas); raw console output in logs/stdout_capture.txt.

[15:32:06.519] [+6241.801s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=4.00, open, 1 vortices, model=:dirac

[15:32:18.884] [+6254.166s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=4.00, open, 1 vortices, model=:dirac

FULL COMPUTATION LOG

[15:32:28.180] [+6263.462s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=4.00, open, 1 vortices, model=:dirac

[15:32:38.165] [+6273.447s] build_ab_cloud_hamiltonian: 96x96 (N=9216), ?=0.0000, t=1.00, W=4.00, open, 1 vortices, model=:dirac

??

TEST 24 (pass 2, HARDCORE): Dirac-string flux ? exhaustive scan

CONSOLE CAPTURE

Full plaquette scan x4 configs: 2 positions x{+1, ?1, +0.3}

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Lattice: 96x96 (N=9216 sites) ? full plaquette scan x4 configs

cfg A (ix=32, iy=48, q=+1.0): vortex 1/1, empty 9024/9024 ? OK

cfg B (ix=64, iy=24, q=+1.0): vortex 1/1, empty 9024/9024 ? OK

cfg A? (ix=32, iy=48, q=-1.0): vortex 1/1, empty 9024/9024 ? OK

cfg A frac (ix=32, iy=48, q=+0.3): vortex 1/1, empty 9024/9024 ? OK

Max |flux| over ALL empty plaquettes and configs: 8.88e-16

??

scan A q=+1.0 PASS ? 9025/9025 plaquettes OK

scan B q=+1.0 PASS ? 9025/9025 plaquettes OK

scan A? q=-1.0 PASS ? 9025/9025 plaquettes OK

scan A frac q=+0.3 PASS ? 9025/9025 plaquettes OK

hermiticity (all) PASS ? verify_hermitian per config

Test 24b [HARDCORE pass 2]: 5 sub-checks, 0 failed ? PASS